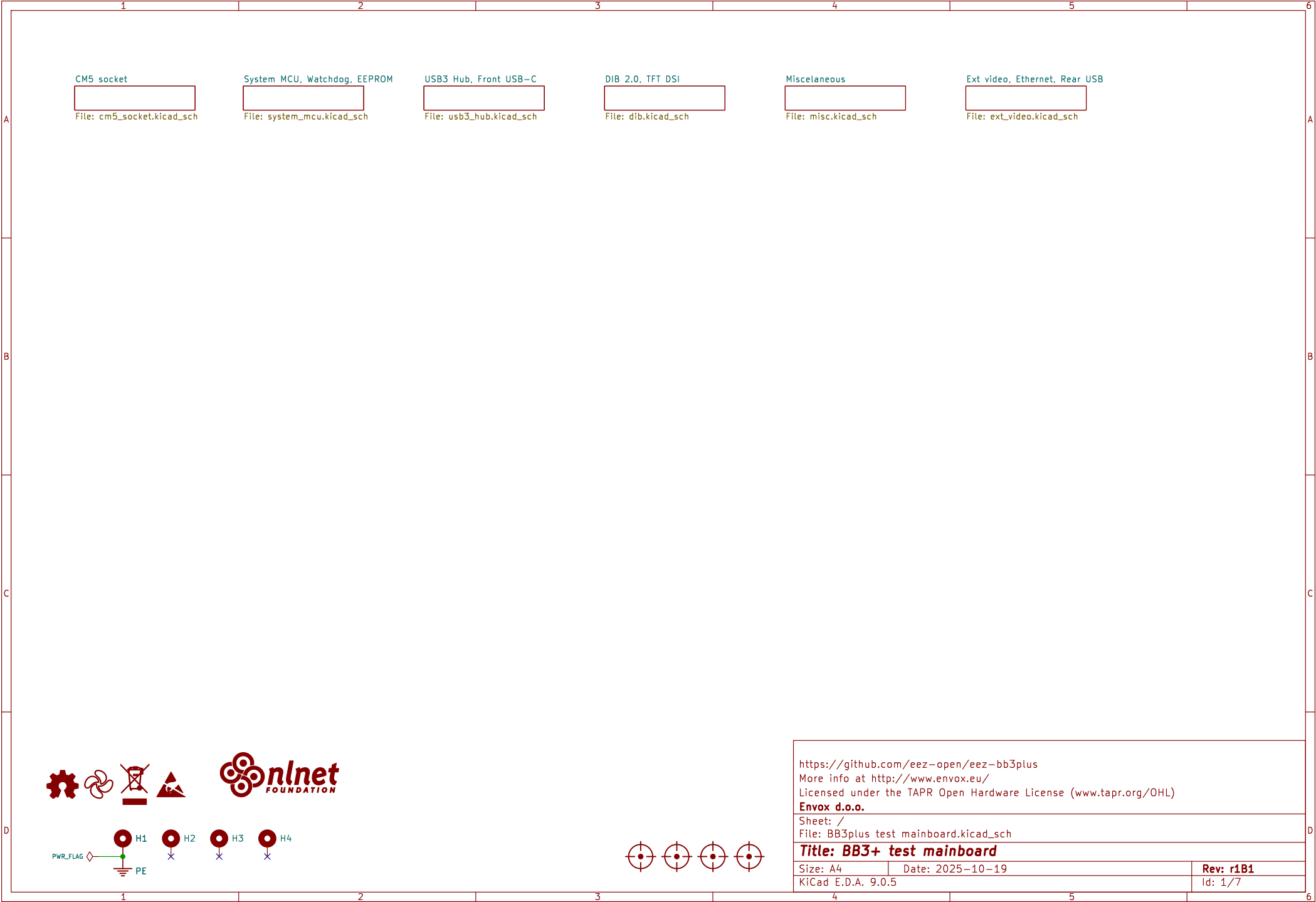
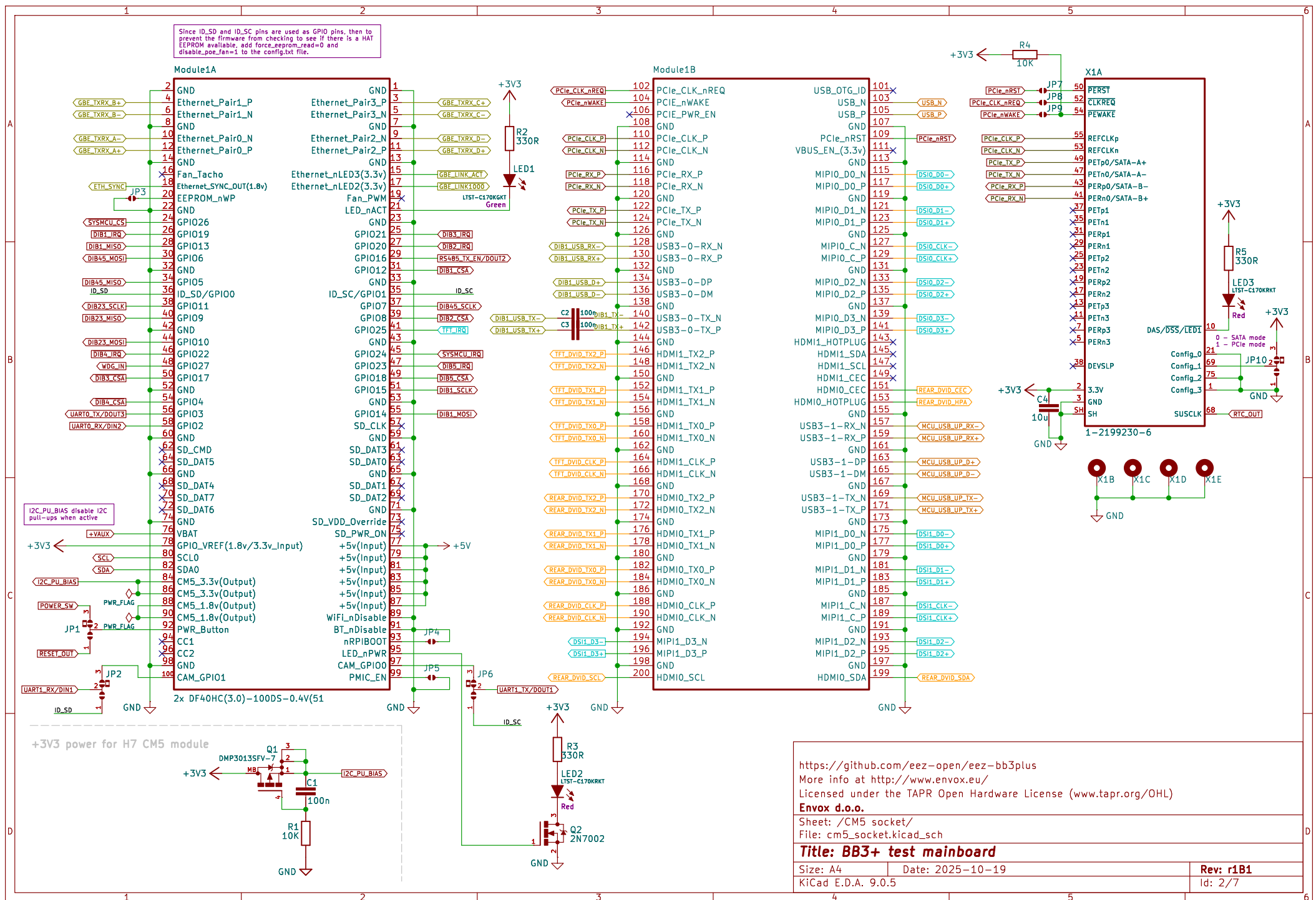


<https://github.com/eez-open/eez-bb3plus>
More info at <http://www.envox.eu/>
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /
File: BB3plus test mainboard.kicad_sch

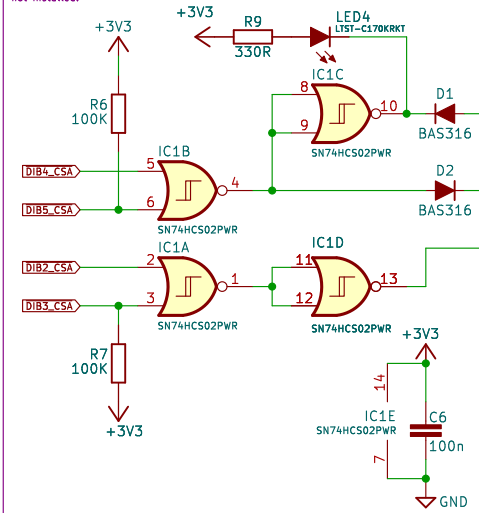
Title: BB3+ test mainboard

Size: A4	Date: 2025-10-19	Rev: r1B1
KiCad E.D.A. 9.0.5		Id: 1/7

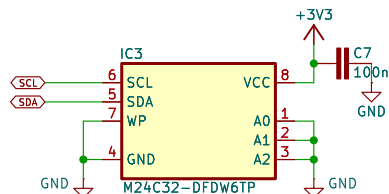


System 32-bit MCU

The System MCU is programmed so that the Master MCU first simultaneously activates DIB4_CSA and DIB5_CSA and then simultaneously activates DIB2_CSA and DIB3_CSA which will reset the System MCU and put it into bootloader mode. Simultaneous activation of two CSA (Chip Select) signals on the same SPI channel should never occur in normal operation. 100K pullup resistors ensure that RESET and BOOT0 are inactive if the Master MCU is not installed.

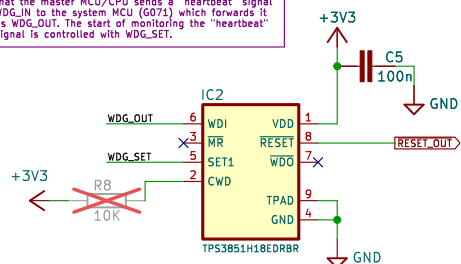


EEPROM

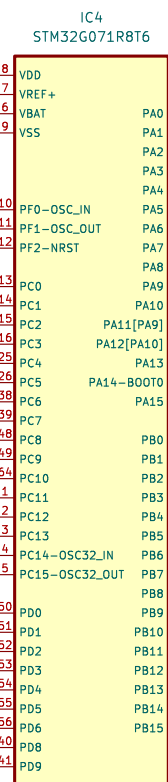
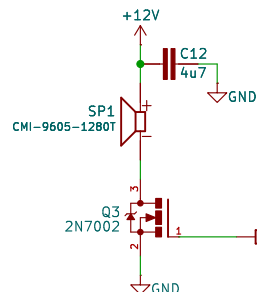


Watchdog/supervisor

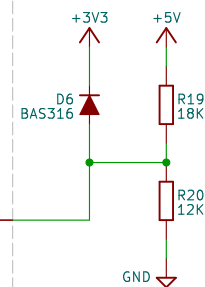
The watchdog signal is daisy chained in such a way that the master MCU/CPU sends a "heartbeat" signal WDG_IN to the system MCU (G071) which forwards it as WDG_OUT. The start of monitoring the "heartbeat" signal is controlled with WDG_SET.



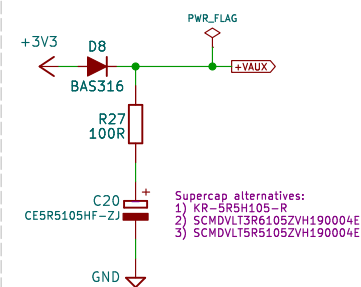
Audio out



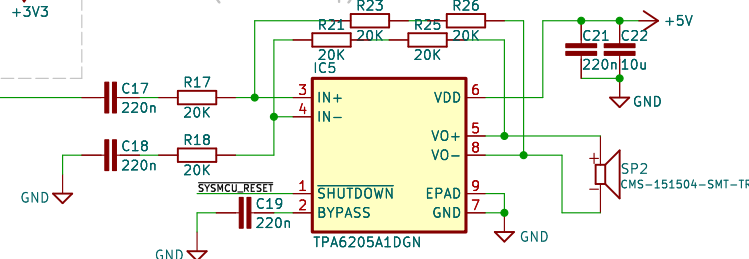
5V monitoring



Supercap for RTC



Audio out (AB-class amp)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /System MCU, Watchdog, EEPROM/

File: system_mcu.kicad_sch

Title: BB3+ test mainboard

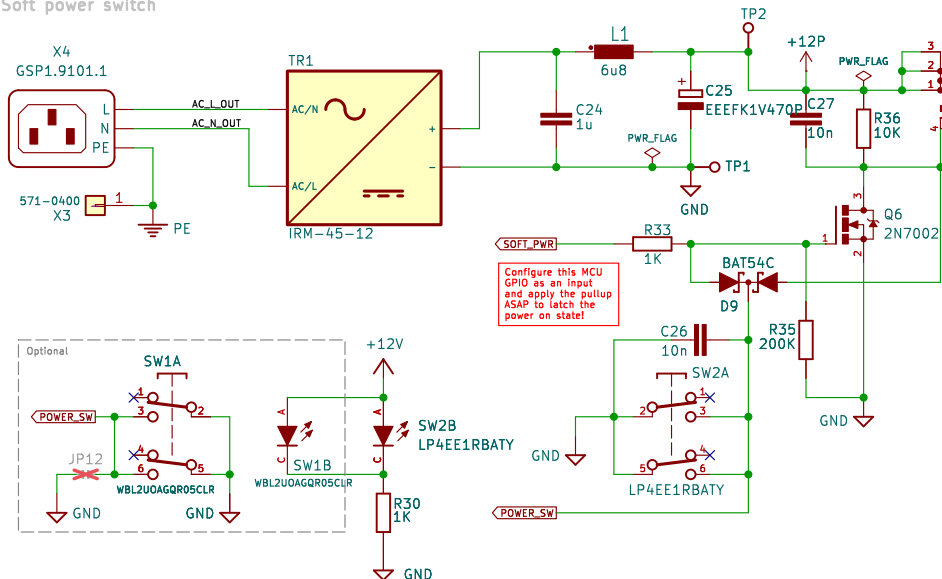
Size: A4 Date: 2025-10-19

KiCad E.D.A. 9.0.5

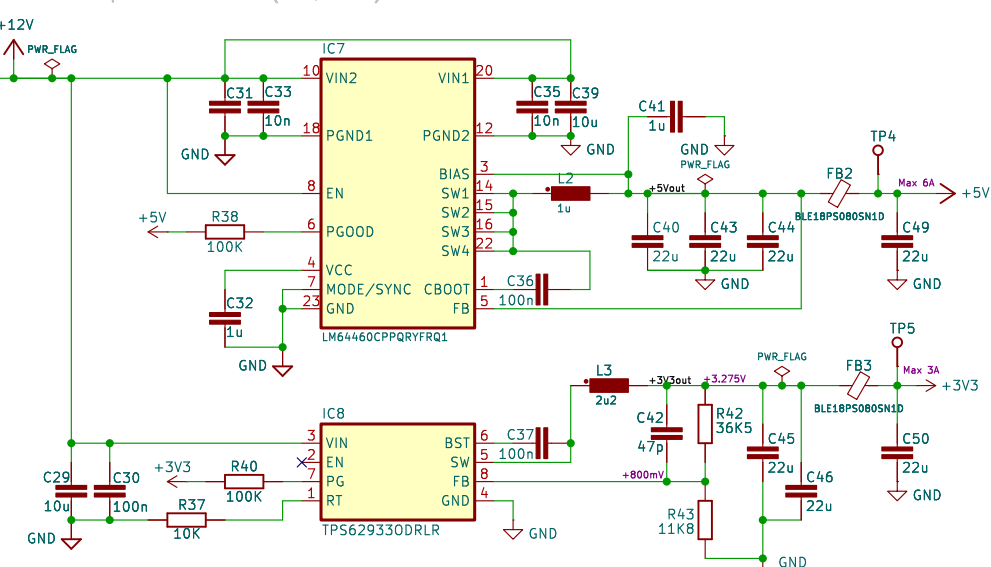
Rev: r1B1

Id: 3/7

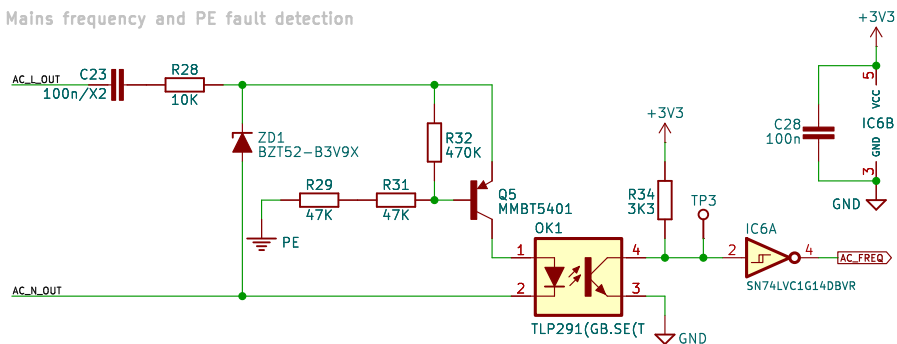
Soft power switch



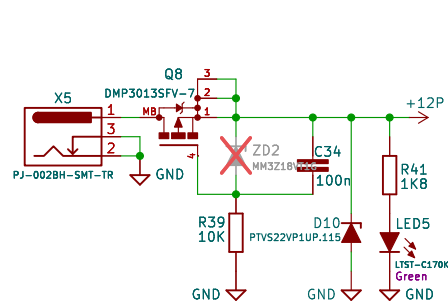
DC-DC stepdown converters (+5V, +3V3)



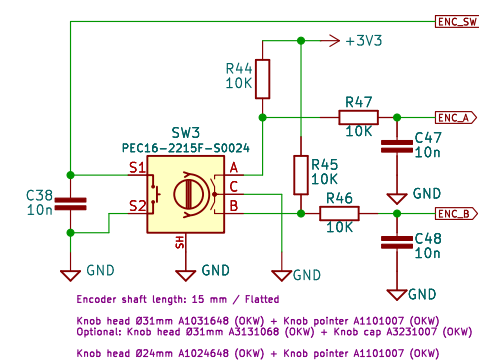
AC Mains frequency and PE fault detection



12Vdc aux



Encoder with switch



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Miscellaneous/

File: misc.kicad_sch

Title: BB3+ test mainboard

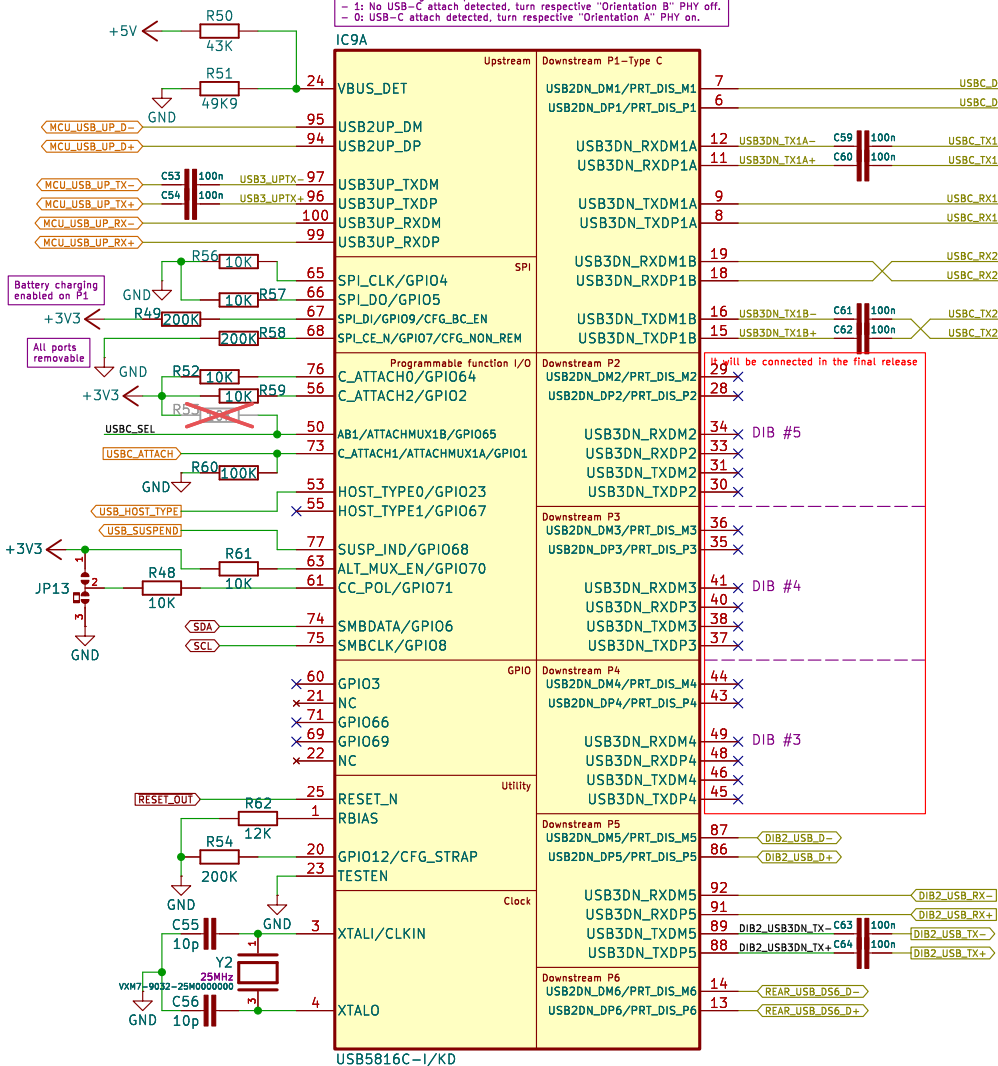
Size: A4 Date: 2025-10-19

KiCad E.D.A. 9.0.5

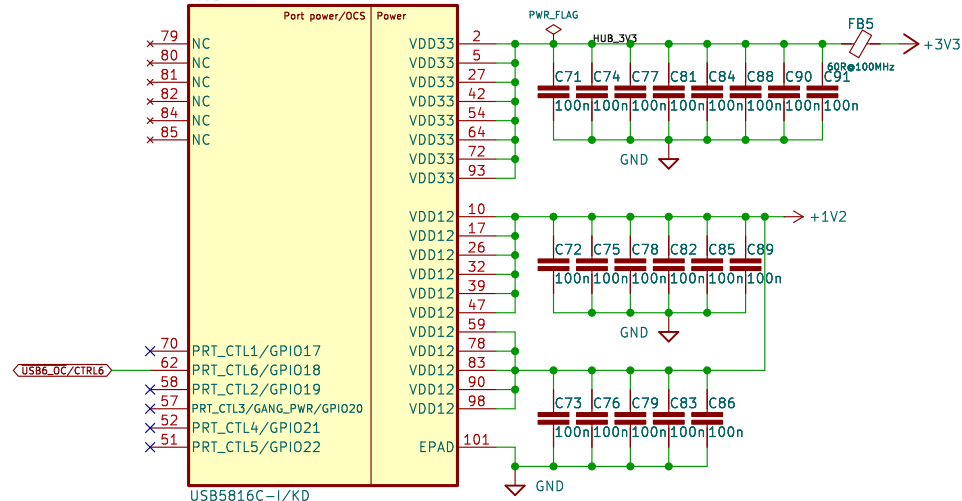
Rev: r1B1

Id: 4/7

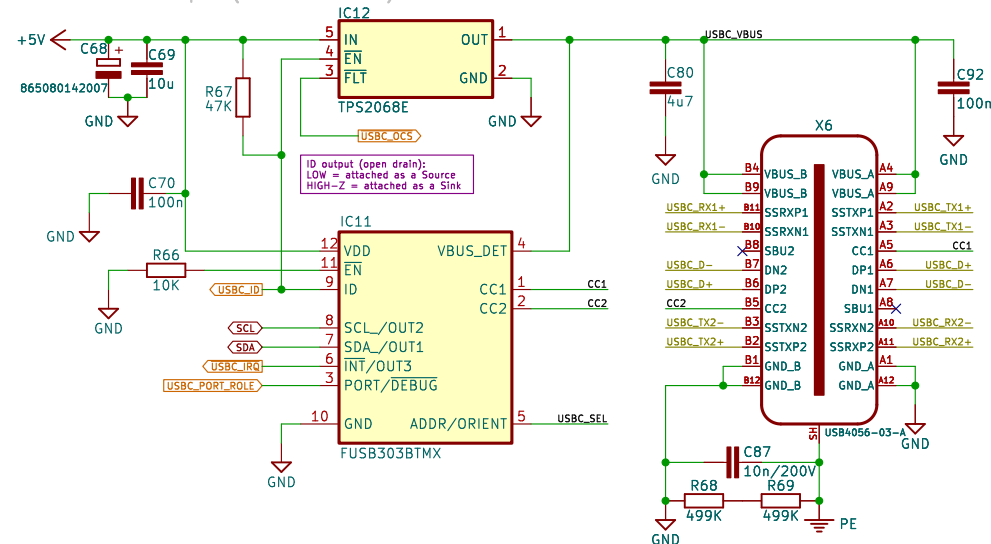
6-port USB hub



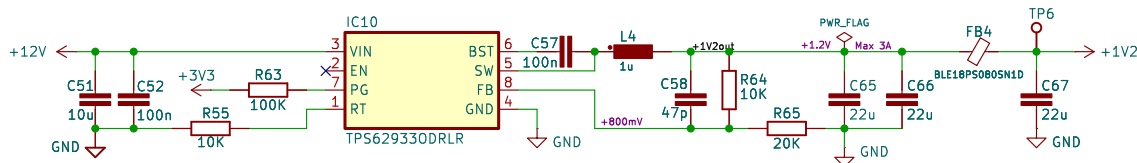
IC9B



USB-C 3.0 front port (source max 1.5A)



USB hub DC-DC stepdown converter (+1V2)



<https://github.com/eez-open/eez-bb3plus>
More info at <http://www.envox.eu/>
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /USB3 Hub, Front USB-C/

File: usb3_hub.kicad_sch

Title: BB3+ test mainboard

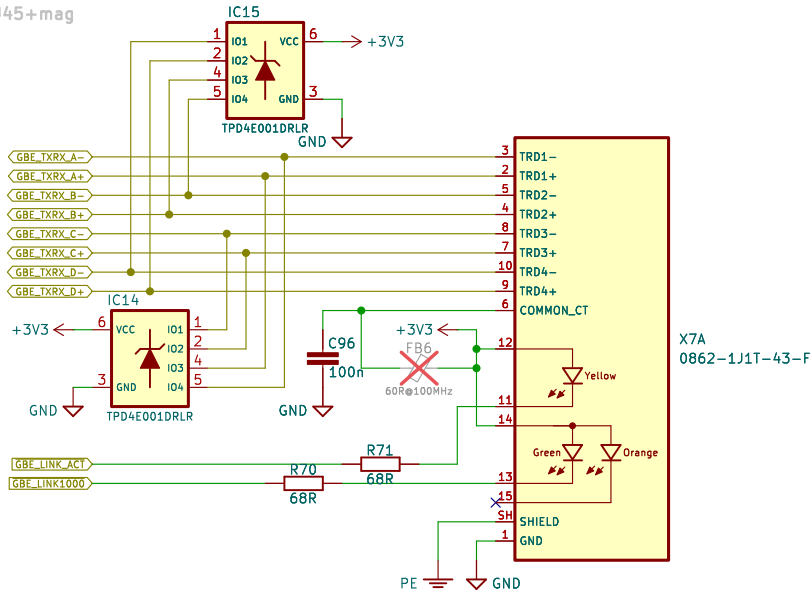
Size: A4	Date: 2025-10-19
----------	------------------

Size: A4	
KiCad E.D.A. 9.0.5	

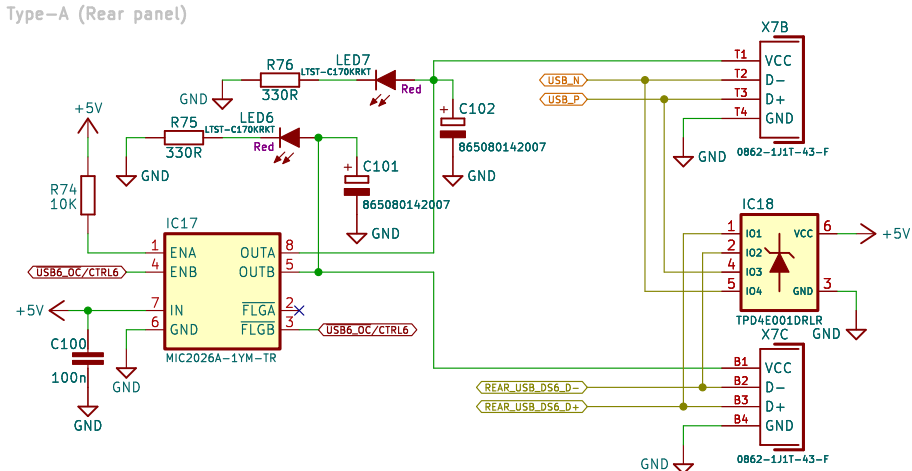
Rev: r1B1

Id: 5/7

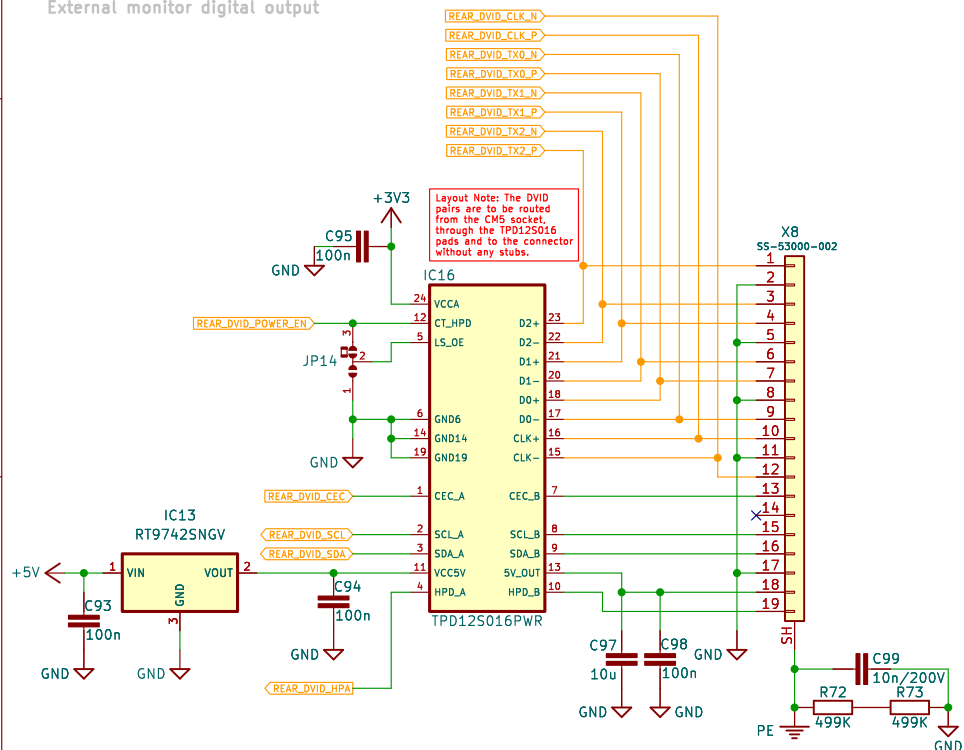
Ethernet RJ45+mag



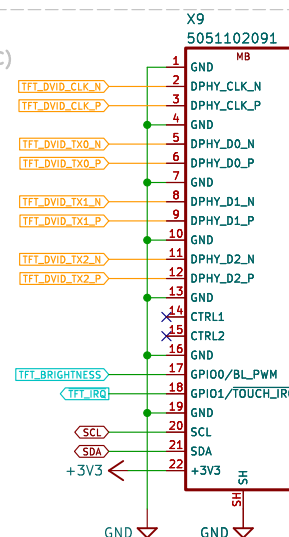
USB2.0 Type-A (Rear panel)



External monitor digital output



TFT Touchscreen display DVID output (22-pin FFC)



<https://github.com/eez-open/eez-bb3plus>
More info at <http://www.envox.eu/>
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /Ext video, Ethernet, Rear USB/
File: ext_video.kicad_sch

Title: BB3+ test mainboard

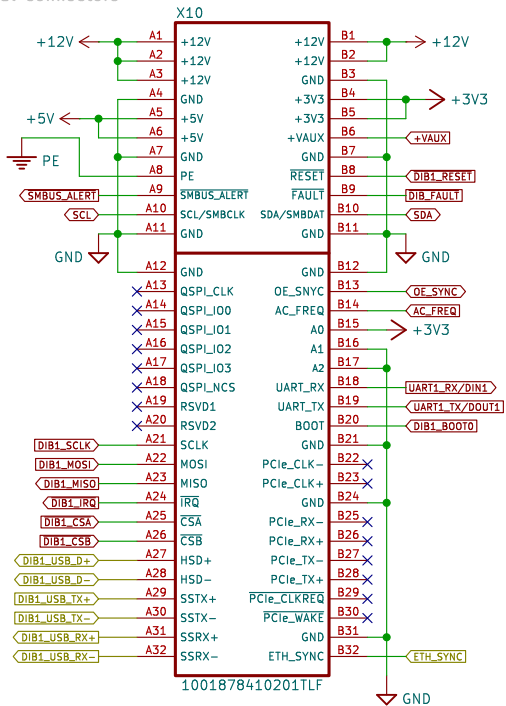
Size: A4 Date: 2025-10-19

KiCad E.D.A. 9.0.5

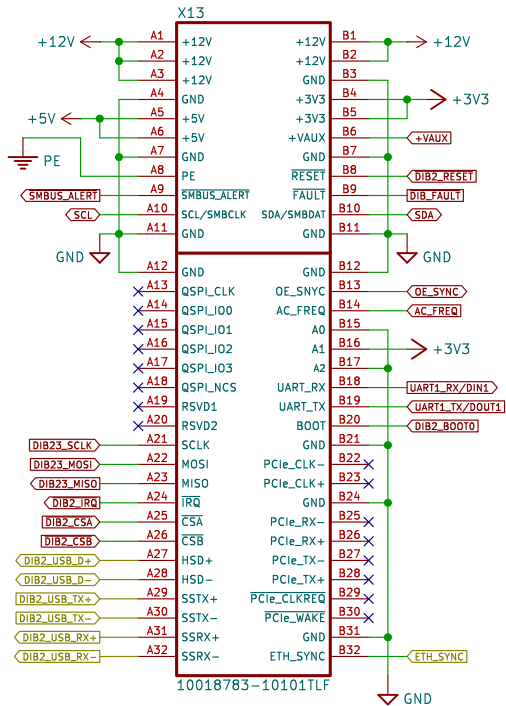
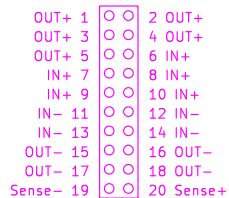
Rev: r1B1

Id: 6/7

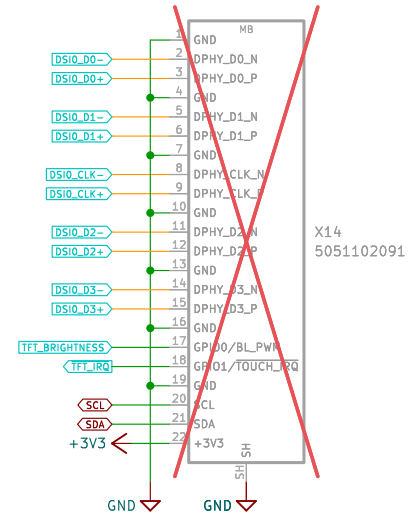
DIB 2.0 Signal connectors



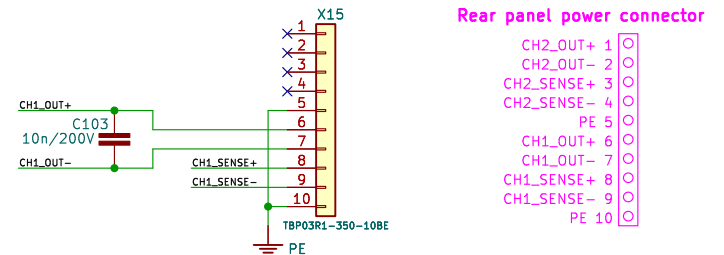
20-pin DIB v2.0 Power connector



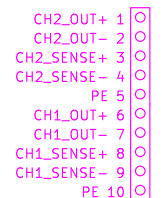
TFT Touchscreen display DSI output (22-pin FFC)



Rear front power outputs



Rear panel power connector



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

Sheet: /DIB 2.0, TFT DSI/

File: dib.kicad_sch

Title: BB3+ test mainboard

Size: A4 Date: 2025-10-19

KiCad E.D.A. 9.0.5

Rev: r1B1

Id: 7/7